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Effects of Zinc Deficiency and Drought on Plant Growth and Metabolism of Reactive Oxygen Species in Maize (*Zea mays* L.)

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Abstract

The combinative effects of applied zinc (Zn) and soil moisture on the plant growth, Zn uptake, and the metabolism of reactive oxygen species (ROS) in maize (*Zea mays* L.) plants were examined through two pot experiments under greenhouse conditions. Maize variety Zhongdan 9409 was used. In experiment 1, maize plants were grown in cumulic cinnamon soil with five Zn treatments (0, 3.0, 9.0, 27.0, and 81.0 mg Zn kg^{-1} soil). Three treatments of soil moisture including serious drought, mild drought, and adequate water supply were set at 30-35%, 40-45%, and 70-75% (w/w) of soil saturated water content, respectively. Soil saturated water content was 36% (w/w). The dry matter weights of shoots were enhanced by Zn application and adequate water supply. There was no apparent difference in plant growth among Zn application rates from 3.0 to 81.0 mg Zn kg^{-1} soil. The increases of plant growth and Zn uptake due to Zn application were found more significant under well-watered condition than under drying condition. In experiment 2, two levels of Zn (0 and 5.0 mg Zn kg^{-1} soil) and soil moisture regimen (40-45% and 70-75% of soil saturated water content, respectively) were set. Zn deficiency or water stress resulted in higher concentrations of $O_2^{\cdot-}$ and malondiadehyde in the first fully expanded leaves. Zn deficiency lowered the activity of superoxide dismutase (SOD, EC 1.15.1.1) in leaves. Drought stress increased SOD activity in leaves regardless of Zn supply. The activity of guaiacol peroxidase (POD, EC1.11.1.11) was found to be enhanced by Zn supply only in well-watered leaves. Zinc deficiency or water stress had little effect on the activity of catalase (CAT, EC 1.11.1.6). The higher ROS level in early maize leaves due to water stress seemed not to be alleviated or lowered partially by Zn application. However, Zn fertilizer was recommended to apply to maize plants irrigated or supplied with adequate water, otherwise Zn deficiency would reduce the water use for plant biomass production.

Key words: drought, maize, reactive oxygen species, zinc deficiency

INTRODUCTION

Zinc (Zn) deficiency is a common nutrient problem in crop production of China. About 60 million ha cropped land is known to be Zn deficient or prone to Zn deficiency, particularly in North China, where soil pH is high and organic matter content is low (Liu *et al.* 1993; Liu 1996). Maize, one of important food crops grown in North China, is sensitive to Zn deficiency. Stunted and chlorotic plants due to Zn deficiency are often observed on early grown maize plants in the field (Liu *et al.* 1993; Liu 1996). Soil drought is the other major constraint to maize growth and productivity in North China, in which annual precipitation is only 200-600 mm (Zhao and Li 1995). Consequently maize plant growth and development are challenged by Zn deficiency and soil moisture deficits concurrently during

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the growing season.

Cakmak *et al.* (1996) suggested that Zn deficiency of plants might be prone to occur in arid and semi-arid regions where soils, particularly topsoil, are usually deficient in water, and so Zn transport from soil solution to roots mostly by diffusion is slight. Wheat yield reductions in Zn deficient calcareous soils were more severe under rainfed than irrigated condition (Ekiz *et al.* 1998). Grewal and Williams (2000) found that adequate Zn nutrition enhanced alfalfa plant growth under the condition of water stress during early vegetative stage. However, Khan *et al.* (2003, 2004) reported that applying Zn increased chickpea grain yields when the plants were well watered, but not under water stress, except for the Zn-efficient and drought-resistant genotype ICC-4958. The effects of Zn nutrition on the responses of maize plants to drought stress and water use of maize plant grown in North China have not received much attention.

Drought results in the increased generation of reactive oxygen species (ROS) due to energy accumulation in stressed plants which absorb more light energy than consumed through photosynthetic carbon fixation (Smirnoff 1993; Asada 1999). Drought inhibits or slows down photosynthetic carbon fixation mainly through limiting the entry of CO_2 into the leaf or directly inhibiting metabolism (Smirnoff 1993; Loggini *et al.* 1999; Apel and Hirt 2004). To counteract the toxicity of ROS, plant cells have developed an antioxidative system, consisting of low-molecular-weight antioxidants like ascorbate, α-tocopherol, glutathione, and carotenoids, as well as protective enzymes. Superoxide radicals are scavenged by superoxide dismutase (SOD), while the resulting H_2O_2 is reduced to H_2O by catalase (CAT) and peroxidase (POD)(Asada 1999; Mittler 2002; Apel and Hirt 2004).

Zn deficiency is also recognized to cause higher levels of ROS in plants and relevant damages to plants (Marschner 1995; Cakmak 2000). Under Zn application, the activity of membrane-bound NADPH oxidase producing ROS decreases (Cakmak and Marschner 1988) and photooxidation process reduces (Marschner and Cakmak 1989), while the activities of SOD, POD, and CAT enhance (Yu *et al.* 1998). Zn plays important roles in lowering the ROS generation and defending cells against ROS attack (Marschner 1995; Cakmak 2000). However, it is not clear whether Zn application can protect plants from the harms of high levels of ROS resulting from drought stress.

Two experiments using Zn-deficient soil from North China were conducted to examine the interactive effects between Zn nutrition and water stress on maize plants. The objectives of this study were to (1) investigate the response of maize plant growth to various Zn application rates under variable soil water supply; (2) determine the effect of soil drought on maize plant growth with different Zn application rates; and (3) reveal whether Zn addition can alleviate ROS damages to maize plants caused by drought stress.

MATERIALS AND METHODS

Experiment 1: The responses of maize plant growth and Zn uptake to various Zn application rates

Tested soil Cumulic cinnamon soil with heavy texture was sampled from Shanxi Province in North China. The soil pH (soil:H_2O as 1:5) was determined as 7.9, organic matter C 8.0 g kg^{-1}, $CaCO_3$ 15.3 g kg^{-1}, NH_4^+-N 15.5 g kg^{-1}, Olsen-P 8.2 g kg^{-1}, NH_4Ac-extractable K 71.5 g kg^{-1}. The concentration of soil DTPA-extractable Zn prior to maize planting was 1.3 mg kg^{-1} below critical soil Zn value. Ten kilograms of air-dried soils passed through a 2-mm nylon sieve were weighed and placed into a poly-ethylene plastic pot with diameter 25 cm (upper) × 18.5 cm (lower) and height 25 cm.

Plant material Maize *cv.* Zhongdan 9409 was selected, which is one of quality protein maize hybrid strains officially released in North China from 1999. The seeds of similar size were surface-sterilized by soaking in 10% (v/v) H_2O_2 for 30 min, washed thoroughly in deionized water and germinated on the filter paper at 25°C for 48 h. Initially, seven uniformly germinated seeds were sown in each pot and thinned to five plants each pot after emergences. After the three-leaf stage, 0.16 g kg^{-1} N, 0.12 g kg^{-1} P_2O_5, and 0.08 g kg^{-1} K_2O were added to soil as solution prepared with urea and KH_2PO_4.

Soil moisture and Zn treatments Four levels of Zn, 3.0, 9.0, 27.0 and 81.0 mg kg^{-1} soil, were applied to soil as solution of $ZnSO_4 \cdot 7H_2O$ before sowing, which were designated as $Zn_{3.0}$, $Zn_{9.0}$, $Zn_{27.0}$ and $Zn_{81.0}$,

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respectively. No zinc addition (Zn_0) was referred as control. Three treatments of soil moisture, 30-35% (serious drought), 40-45% (mild drought), and 70-75% (adequate water supply), were imposed from four-leaf stage till harvest, which were represented by symbols W_1, W_2, and W_3, respectively. Soil saturated water content was 36% (w/w). Soil moisture was controlled by weight measurement method every day. If the change in soil moisture was 10-20 g kg^{-1}, deionized water was added. To minimize the effects of differences in plant size induced by the Zn treatments on the development of water stress, Time Domain Reflectometry (TRASE, Soil Moisture Equipment Corp., USA) with 15 cm length waveguides was used to measure soil moisture content. The soil bulk density in pots was 1.70 g cm^{-3} estimated by the ratio of soil weight to soil volume in a pot.

Harvest and chemical analysis The experiment was conducted in a glasshouse. Temperature within the greenhouse was 30±3°C during the day and 17±3°C during the night. Plants were grown under natural day length. After growth for 63 days, plants were harvested and separated into roots, leaves, and stems. Soils were washed out of the roots. These plant samples were rinsed in distilled water, killed at 105°C for 30 min, dried at 65-70°C for 72 h and weighed as biomass. Dry plant materials were digested in HNO_3-$HClO_4$ and the extract was analyzed for Zn using an atomic absorption spectrophotometer.

Experiment 2: Effects of Zn on ROS metabolism in plants under soil-dried condition

Plant growing conditions The same soil and maize cultivar as experiment 1 were used in experiment 2. Four kilograms of air-dried soils were placed into a poly-ethylene plastic pot with 14.6 cm diameter and 20 cm height. Two Zn treatments of 0 and 5.0 mg kg^{-1} soil (applied as $ZnSO_4 \cdot 7H_2O$ solution before sowing) were designated as Zn_0 and $Zn_{5.0}$, respectively. Two soil moisture treatments including 40-45% (drought) and 70-75% (well-watered condition) of soil saturate water content, respectively, were imposed at the four-leaf stage till harvest.

Experiment 2 was conducted in the same glasshouse as experiment 1. 0.16 g kg^{-1} N, 0.12 g kg^{-1} P_2O_5, and 0.08 g kg^{-1} K_2O were added to soil as solution prepared with urea and KH_2PO_4 at the three-leaf stage of maize plant.

At 35 day after planting, the fully expanded young leaves were collected at 9:00-9:30 a.m., immediately frozen in liquid nitrogen (-80°C), and stored until analysis.

Enzyme extraction and activity determination One to two grams of leaves were ground with a pestle in an ice-cold mortar with 10 mL of 0.05 mol L^{-1} sodium phosphate buffer (pH 7.8). The homogenates were filtered through four layers of cheese cloth and then centrifuged at 4°C for 20 min at 15 000×g. The supernatants were stored at 4°C for determination of $O_2^{\cdot-}$ and enzyme activities.

The SOD activity was measured spectrophotometrically according to its ability to inhibit the photochemical reduction of nitroblue tetrazolium (NBT) (Beyer and Fridovich 1987). One unit of SOD was defined as the amount required to inhibit the photoreduction of nitroblue tetrazolium by 50%. The specific activity of SOD was expressed as units (U) mg^{-1} protein.

The POD activity was determined by the method of Chance and Maehly (1955) using guaiacol as an electron donor. The CAT activity was assayed by the method of Clairborne (1985). The decomposition of H_2O_2 was followed by a decline in absorbance at 240 nm for 2 min. The activities of POD and CAT were expressed as the changes of absorbance (ΔA) in one minute per mg protein.

The protein content in the supernatant was estimated based on Bradford's method (1976), using bovine serum albumin as standard.

The $O_2^{\cdot-}$ determination The above extract was used to determine the concentration of $O_2^{\cdot-}$ based on hydroxylamine oxidation reaction $NH_2OH + 2O_2^{\cdot-} + H_2O \rightarrow -NO_2^{\cdot-} + 2H_2O_2$. The production of NO_2^- was quantified using colorimetry method (Wang and Luo 1990).

The determination of malondiadehyde (MDA) MDA is a decomposition product from the peroxidation of polyunsaturated fatty acids. One gram of leaves was ground in a mortar with liquid nitrogen and extracted with 10 mL 10% (v/v) trichloroacetic acid (TCA). After centrifuging at 4 000×g for 10 min at 4°C, supernatants were pooled and an aliquot of appropriately diluted

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sample was added to a test tube to react with 0.6% (w/v) thiobarbituric acid (TBA) solution comprised of 10% (v/v) TCA. Samples were heated at 95°C for 15 min, and after cooling, absorbances were read at 450, 532, and 600 nm. MDA equivalents (nmol mL^{-1}) were calculated as $6.45 \times (A_{532}-A_{600}) - 0.56 \times A_{450}$. MDA concentration in leaves was estimated per fresh leaf mass (Zhu *et al.* 1990).

Statistics analysis Each experiment was set up in a completely randomized factorial design with three replicates. All data were subjected to the analysis of variance (ANOVA), including separation of main effects Zn supply and soil moisture treatments and their interactive effects.

RESULTS

Experiment 1

Dry matter production After 5 to 6 leaf-stage, Zn deficiency symptoms such as chlorosis followed by white necrotic spots appeared in young leaves. Zn deficiency symptoms were first observed in the leaf blades with adequate water supply, but were not obvious in plants with low growth when subjected to serious drought stress.

Soil water shortage reduced the biomass of roots, stems, and leaves irrespective of Zn application rates (Table 1). Serious drought stressed (W_1) plants had the lowest dry weights of roots, stems, and leaves among the three treatments of soil moisture. The reduction of dry matter due to drought stress was greater in Zn applied plants compared with Zn_0 plants.

The interactive effect between Zn supply and soil moisture treatments on plant growth was significant. Zn supply increased substantially the dry matter weight of stems and leaves under well-watered condition, but not under soil-dried condition. Zn deficiency decreased the water use for biomass production of maize plants. The largest enhancement on biomass by Zn nutrition improvement was in stems, next in leaves, and the smallest was in roots. There was no significant difference in dry matter weights among four Zn application rates from 3.0 to 81.0 mg kg^{-1} soil (Table 1), which was similar to the results of Liu *et al.* (1993).

The same results were also observed in experiment 2 (data not shown).

Zn concentration and Zn content The Zn concentration in roots, stems and leaves increased with increasing of Zn application rates. The Zn concentration in leaves at Zn_0, $Zn_{3.0}$, and $Zn_{9.0}$ was lower under serious drought than under mild drought and well-watered condition, which may account for the low plant growth and Zn translocation from roots to shoots caused by serious drought. More notable increases of Zn concentration in roots, leaves, and stems resulted from Zn application were observed under drought compared with adequate water supply (Table 2).

• Zn content in leaves, stems, and roots increased with increasing of Zn application rates. Stems and leaves accumulated more Zn than roots. As Zn supply increased, Zn accumulation in shoots (stems and leaves) was higher than that in roots, indicating Zn application could enhance Zn translocation from roots to shoots.

Zn content in roots, stems, and leaves declined significantly with soil drought irrespective of Zn application rates. Compared with adequate water supply, serious and mild drought stresses led to a more decrease of Zn content in leaves, stems, and roots (Table 3).

Table 1 Effects of Zn and soil moisture on dry matter weights (g/pot) of roots, stems, leaves, and total plants

Treatments	Roots				Stems				Leaves				Total plants			
	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean
Zn_0	1.44	2.89	3.42	2.58	0.56	1.01	2.85	1.48	4.71	10.69	19.74	11.71	6.71	14.60	26.01	15.77
$Zn_{3.0}$	1.80	2.95	4.77	3.17	0.77	2.01	3.96	2.25	5.84	13.67	28.25	15.92	8.42	18.63	36.98	21.34
$Zn_{9.0}$	1.62	2.96	4.42	3.00	1.04	2.63	4.65	2.77	5.42	14.30	28.59	16.10	8.09	19.89	37.66	21.88
$Zn_{27.0}$	1.63	2.85	4.23	2.91	0.66	2.60	5.00	2.75	5.03	14.88	28.57	16.16	7.32	20.34	37.80	21.82
$Zn_{81.0}$	1.47	2.85	4.13	2.82	0.82	2.00	4.32	2.38	4.82	12.33	25.31	14.15	7.12	17.17	33.76	19.35
Mean	1.59	2.90	4.19	2.90	0.77	2.05	4.16	2.33	5.17	13.17	26.09	14.81	7.53	18.12	34.44	20.03
LSD (P=0.05)																
Zn	0.19				0.25				0.82				1.02			
Water	0.15				0.19				0.63				0.79			
Zn × Water	0.33				0.43				1.47				1.76			

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Experiment 2

The concentration of $O_2^{\cdot-}$ and MDA in leaves The levels of $O_2^{\cdot-}$ in leaves were increased by drought stress. The highest $O_2^{\cdot-}$ value was observed in leaves, stressed by Zn deficiency combined with drought. Zn application reduced $O_2^{\cdot-}$ production in water-stressed leaves, but did not in well-watered leaves.

Drought stress and Zn deficiency led to high MDA concentration in leaves, but they did not show positively interactive effect on MDA concentration (Table 4).

Activities of some enzymes scavenging ROS Zn deficiency did not significantly decrease SOD activity in leaves under drought-stressed condition, while resulted in a much greater reduction (52.3%) under well-watered condition.

Drying condition induced increase of SOD activity in leaves, but the increase of SOD activity was larger in Zn deficient plants than that with Zn application.

The activities of POD in well-watered leaves were enhanced by Zn supply, but not being changed in water-stressed leaves. There was no notable difference of CAT activities among the treatments with Zn and soil water (Table 5).

Table 2 Effects of Zn application and soil moisture supply on Zn concentration (μg g^{-1} dry matter) of roots, stems, and leaves

Treatments	Roots				Stems				Leaves			
	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean
Zn_0	38.3	19.7	25.0	27.7	37.3	27.2	26.7	30.4	10.3	13.2	13.6	12.4
$Zn_{3.0}$	53.5	40.5	33.4	42.4	79.1	71.8	49.6	66.8	24.3	30.4	30.4	28.4
$Zn_{9.0}$	57.0	42.4	36.1	45.2	139.7	113.3	66.9	106.7	33.3	36.5	39.2	36.3
$Zn_{27.0}$	81.9	60.9	57.1	66.6	246.9	170.5	119.0	178.8	50.3	50.5	49.2	50.0
$Zn_{81.0}$	168.3	137.4	116.7	140.8	337.8	235.3	195.9	256.4	69.1	85.0	61.7	71.9
Mean	79.8	60.2	53.7	64.5	168.2	123.6	91.6	127.8	37.5	43.1	38.8	39.8
LSD (P=0.05)												
Zn	3.4	8.6	3.5									
Water	1.4	6.7	2.7									
Zn × Water	5.9	14.9	6.1									

Table 3 Effects of Zn application and soil moisture supply on Zn content (mg/pot) of roots, stems, and leaves

Treatments	Roots				Stems				Leaves			
	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean	W_1	W_2	W_3	Mean
Zn_0	54.7	57.1	85.8	65.9	21.0	27.3	76.8	41.7	48.07	141.7	270.1	153.3
$Zn_{3.0}$	96.4	119.5	158.9	124.9	61.1	144.2	196.3	133.9	141.82	412.5	860.1	471.5
$Zn_{9.0}$	92.8	125.8	159.5	126.0	144.5	297.5	311.4	251.1	180.48	521.4	1 121.5	607.8
$Zn_{27.0}$	133.3	174.0	241.5	182.9	163.5	443.1	594.9	400.5	250.13	748.8	1 406.0	801.7
$Zn_{81.0}$	248.0	390.9	482.7	373.9	276.6	469.7	847.8	531.3	332.43	1 049.7	1 560.3	980.8
Mean	125.0	173.5	225.7	174.7	133.3	276.3	405.4	271.7	190.59	574.8	1 043.6	603.0
LSD (P=0.05)												
Zn	18.9				33.3				58.0			
Water	14.6				25.8				44.9			
Zn × Water	32.8				57.7				93.6			

Table 4 The concentrations of $O_2^{\cdot-}$ and MDA in leaves of maize plants influenced by Zn supply and soil moisture conditions

Treatments		$O_2^{\cdot-}$ concentration (mmol min^{-1} g^{-1} FW)	MDA concentration (mmol g^{-1} FW)
Zn_0	Drought	1.50 ± 0.16 c	7.86 ± 0.30 d
$Zn_{5.0}$	Drought	1.25 ± 0.12 b	6.81 ± 0.45 c
Zn_0	Well-watered	1.10 ± 0.06 ab	5.54 ± 0.24 b
$Zn_{5.0}$	Well-watered	0.89 ± 0.10 a	4.63 ± 0.09 a

Data are the means ± SD. The data within a column followed by the same letter are not significantly different at P=0.05.

Table 5 Activities of SOD, POD and CAT in leaves of maize plants influenced by Zn supply and soil moisture conditions

Treatments		SOD activity (U mg^{-1} protein)	POD activity (ΔA mg^{-1} protein min^{-1})	CAT activity (ΔA mg^{-1} protein min^{-1})
Zn_0	Drought	38.49 ± 1.89 bc	7.33 ± 0.75 a	2.13 ± 1.02 a
$Zn_{5.0}$	Drought	41.45 ± 2.98 c	7.17 ± 0.75 a	2.46 ± 0.37 a
Zn_0	Well-watered	23.82 ± 3.46 a	7.21 ± 0.86 a	2.10 ± 0.28 a
$Zn_{5.0}$	Well-watered	36.28 ± 0.39 b	9.64 ± 1.08 b	1.57 ± 0.19 a

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DISCUSSION

Zn nutrition of plants is considered one of major problems in the arid and semi-arid regions (Ekiz *et al.* 1998). Wheat yield reductions in Zn deficient calcareous soils were greater under rainfed than irrigated conditions (Ekiz *et al.* 1998). Grewal and Williams (2000) found that, at low Zn supply, soil moisture stress significantly depressed shoot dry matter weight and leaf area in alfalfa plants, while there was little impact of soil moisture treatments when supplied Zn concentration was high. However, the present studies with two pot experiments showed that imposing water stress reduced biomass production of maize cultivar Zhongdan 9409 regardless of Zn fertilization. Zn application, even with higher rates above 3.0 mg kg^{-1} soil, seemed not to enhance the biomass of water-stressed plants, but increased the dry matter weight of well-watered plants. Similar results were revealed by Khan *et al.* (2003, 2004) who found that chickpea yield losses from water stress were the greatest at the highest level of Zn. They attributed the possible reasons to the limited soil volume afforded by the pots and the rapid development of stress on the larger plants grown at adequate levels of Zn.

The lower diffusion of Zn in dry soil restricts uptake of Zn and may exacerbate Zn deficiency (Ekiz *et al.* 1998). However, the present experiment showed that the Zn concentrations in various plant parts without Zn application were not increased with soil moisture supply and were still below the generally critical level (about 20 μg g^{-1}) of plant Zn deficiency (Liu *et al.* 1993; Liu 1996; Marschner 1995), even though maize plants were well watered. Visual Zn deficiency symptoms were not only observed in plants under mild soil drought, but also in those with adequate soil water supply. In calcareous soils after the flush in microbial activity that follows submergence, HCO_3^- is the predominant anion, which mainly decreases Zn translocation from root to shoot and to a lesser extent, Zn uptake by roots (Marschner 1995; Dobermann and Fairhurst 2000; Hajiboland *et al.* 2005). With adequate soil moisture regimes, HCO_3^- concentration in soil solution may increase, and thereby result in more marked response of maize plant growth to Zn application, which needs further determination.

It was assumed that there are different complicated mechanisms by which soil moisture content influences plant Zn uptake and utilization besides the influence on the diffusion rate of Zn ions in soils. First, root growth is most commonly reduced by soil drying, but is usually less inhibited than shoot growth, and sometimes may even be promoted, resulting in an increase of root-to-shoot ratio under limited water content of soils (Davies and Bacon 2003; Wang and Yamauchi 2006). Corresponding enhancement of root uptake Zn by relatively large root biomass under drought remains unclear. Second, the stimulation of mucilage secretion in response to increased mechanical impedance in dry soil (Czarnes *et al.* 2000; Wang and Yamauchi 2006) is suggested to contribute to the maintenance of Zn uptake by facilitating Zn^{2+} transport from embedded soil particles to the root surface (Nambiar 1976a, b). This process may be further promoted by a hydraulic lift that root transfers water from the subsoil and subsequently release into the dry top soil layer (Vetterlein and Marschner 1993). Third, soil drought may decrease the demand for nutrients by plants due to their low growth. Fourth, Zn translocation from roots to shoots is inhibited by low water supply to plants. The possible relationship between plant Zn nutrition and drought stress should be further determined.

Zn deficiency results in an increase of ROS level, which disturbs plant growth. Zn is suggested to play an important role in protecting plant cells from the damage of ROS (Marschner 1995; Cakmak 2000). Thus it would raise a question that whether the improvement of Zn nutritional status of plants is of great importance for their survival from the damage of ROS under drought-stressed condition. The present study did not find such points with the maize cultivar Zhongdan 9409. Drought stress or Zn deficiency resulted in higher concentrations of $O_2^{\cdot-}$ and MDA in leaves, but they did not show a significantly positive interaction. SOD activity in leaves was lowered by Zn deficiency, but it was increased by drought stress. The activity of POD was found to be enhanced by Zn supply only in the leaves with well-water supply. Similar results were shown by Strack *et al.* (1996) and Casona *et al.* (1997) that, irrespective of the Zn nutritional status of plants, the activity of CuZn-SOD, one form of SOD, was decreased when plants were exposed to a strong photo-oxidative stress or the conditions

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producing high amounts of H_2O_2.

The interesting results were shown in some Zn-hyperaccumulator species such as *Alyssum murale* and *Thlaspi caerulescens* (*Brassicaceae*), Zn hyperaccumulation did not ameliorate the inhibition of growth by water stress since Zn hyperaccumulation had minimal effects on the osmolality of leaf-sap extracts, relative water content of the shoots, or rate of evapotranspiration. Moreover, water stress did not induce great changes in shoot metal concentrations of these constitutive hyperaccumulators (Whiting *et al.* 2003).

In conclusion, there seems to be no deleterious effect from the applied Zn under a dry weather. Although Zn application is less useful to enhance maize Zhongdan 9409 tolerance to drought stress during early growth, Zn supply is recommended to increase the water use for biomass production if plants are well watered. But these conclusions can be reached only in the context of pot studies in greenhouse, and should be validated in the field.

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